Bayesian additive regression trees

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1 Introduction

Bayesian Additive Regression Trees (BART), introduced by Chipman et al. [2010], is a nonparametric Bayesian approach to supervised learning designed to estimate an unknown regression function from data. Instead of specifying a parametric form for f, BART approximates it using an ensemble of regression trees whose predictions are combined additively.

The key idea underlying BART is to represent the regression function as a sum of many simple tree-based models:

$$f(x) \approx \sum_{j=1}^{m} g(x; T_j, M_j),$$

where each $g(x; T_j, M_j)$ denotes a regression tree with structure T_j and terminal node parameters M_j. Each tree acts as a weak learner that explains only a small portion of the signal, and the final prediction is obtained by summing the contributions of all trees.

BART can be understood as a Bayesian counterpart to ensemble tree methods such as boosting, bagging, and random forests. Like boosting, it constructs an additive model of trees, however, rather than sequentially fitting trees using optimization, BART formulates a full probabilistic model with a prior distribution on tree structures and parameters. Inference is then performed using a Markov chain Monte Carlo (MCMC) algorithm, which iteratively updates each tree conditioned on the others.

Beyond its flexibility, BART offers several practical advantages. Its Bayesian formulation provides natural uncertainty quantification through the posterior, while the use of regularization priors makes the method largely insensitive to hyperparameter tuning. Empirically, BART achieves predictive performance competitive with popular machine learning baselines such as gradient boosting, random forests, neural networks, and regularized regression methods like the Lasso across a wide range of datasets.

In this project, we first present the BART algorithm in Section 2. We then experimentally evaluate its performance in Section 3. In particular, we illustrate its ability to quantify epistemic uncertainty, to perform variable selection, and to handle high-dimensional data. We complement the experimental study of Chipman et al. [2010] by considering additional datasets and more recent baselines, such as XGBoost Chen and Guestrin [2016] and LightGBM Ke et al. [2017], which were not yet available when BART was originally introduced.

2 The BART Algorithm

2.1 Model

BART models the unknown regression function using an additive ensemble of regression trees. Given predictors $x \in \mathbb{R}^p$ and response Y, the model assumes

$$Y = f(x) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2),$$

where the function f is approximated as a sum of m regression trees:

$$f(x) = \sum_{j=1}^{m} g(x; T_j, M_j).$$

Here T_j denotes the structure of the j-th tree and M_j the set of values associated with its terminal nodes. Each tree partitions the predictor space using binary splitting rules of the form $x_i \leq c$ versus $x_i > c$. Observations are routed through the tree according to these rules until reaching a terminal node, whose associated parameter determines the prediction contribution of that tree.

The ensemble therefore combines many simple trees, each capturing a small portion of the signal. Because trees may involve different variables and depths, the resulting model can represent both nonlinear effects and complex interactions between predictors.

2.2 Prior

BART places a prior distribution over the tree structures, the terminal node parameters, and the noise variance. The prior is factorized as

$$p((T_1, M_1), \ldots, (T_m, M_m), \sigma) = \left(\prod_{j=1}^{m} p(T_j) p(M_j | T_j) \right) p(\sigma).$$

where $g(x; T_j, M_j)$ denotes a regression tree with structure T_j and terminal node parameters M_j.

The prior over tree structures $p(T_j)$ favors shallow trees by assigning a decreasing probability of splitting as the depth d of a node increases. More precisely, a node at depth d splits with probability

$$\alpha(1 + d)^{-\beta},$$

where $\alpha \in (0, 1)$ and $\beta \geq 0$ control the expected tree size. Splitting variables and split points are typically chosen uniformly among the available predictors and candidate cutpoints.

For the terminal node parameters, each leaf value μ_{ij} is assigned an independent Gaussian prior

$$\mu_{ij} \sim \mathcal{N}(0, \sigma_\mu^2),$$

with σ_μ chosen to shrink individual tree contributions toward zero. Because the regression function is the sum of m trees, this prior ensures that each tree explains only a small portion of the signal.

Finally, the noise variance is given a conjugate inverse–chi-square prior

$$\sigma^2 \sim \nu\lambda / \chi_\nu^2,$$

Together, these priors regularize the model and prevent individual trees from dominating the ensemble.

2.3 Posterior Sampling via MCMC

BART uses a Markov Chain Monte Carlo algorithm based on Gibbs sampling.

At each iteration, the algorithm updates the m trees sequentially while conditioning on the others. For the j-th tree, we first compute the *partial residuals*

$$R_{ij} = y_i - \sum_{k \neq j} g(x_i; T_k, M_k),$$

which represent the portion of the response not explained by the other trees. We observe that the update of (T_j, M_j) is equivalent to fitting a single regression tree to the dataset $\{(x_i, R_{ij})\}_{i=1}^n$.

Sampling the tree parameters is performed in two steps. First, a new tree structure T_j is proposed using a Metropolis–Hastings step that locally modifies the current tree through operations such as growing a leaf, pruning a branch, or changing a split rule. The acceptance probability depends on the posterior ratio

$$p(T_j \mid R_j, \sigma) \propto p(T_j)\, p(R_j \mid T_j, \sigma),$$

where $p(R_j \mid T_j, \sigma)$ is obtained by integrating out the leaf parameters under their Gaussian prior.

Once a new structure T_j is accepted, the terminal node parameters are sampled independently for each leaf. If leaf ℓ contains the observations $i \in I_\ell$, the posterior distribution of its value μ_ℓ is Gaussian,

$$\mu_\ell \mid T_j, R_j, \sigma \sim \mathcal{N}(m_\ell, v_\ell),$$

where $n_\ell = |I_\ell|$ and

$$v_\ell = \left(\frac{n_\ell}{\sigma^2} + \frac{1}{\sigma_\mu^2} \right)^{-1}, \qquad m_\ell = v_\ell \frac{\sum_{i \in I_\ell} R_{ij}}{\sigma^2}.$$

After all trees have been updated, the noise variance is sampled from its conditional posterior. Let

$$\epsilon_i = y_i - \sum_{j=1}^{m} g(x_i; T_j, M_j)$$

be the current residuals. With the inverse–chi-square prior $\sigma^2 \sim \nu\lambda/\chi^2_\nu$, the posterior remains conjugate:

$$\sigma^2 \mid \epsilon \sim \frac{\nu\lambda + \sum_{i=1}^{n} \epsilon_i^2}{\chi^2_{\nu+n}}.$$

The algorithm cycles through these updates repeatedly, generating a Markov chain whose stationary distribution is the posterior over the model parameters.

2.4 Inference

For inference, we consider the average of the n_{samples} MCMC iterations obtained after discarding the first n_{burn} iterations used for mixing (burn-in). The estimator is therefore

$$\hat{f}(x) = \frac{1}{n_{\text{samples}}} \sum_{s=1}^{n_{\text{samples}}} f^{(s)}(x),$$

where $f^{(s)}(x)$ denotes the prediction obtained at the s-th MCMC iteration after burn-in.

3 Experiments

For our experiments, we use the Python implementation of BART provided by the `bartpy` library Coltman [2018]. Our code is available at the following GitHub repository: `https://github.com/loumock/project_MVA_Bay`. For our experiments, we do not perform any specific hyperparameter tuning, neither for the BART prior nor for the hyperparameters of the baseline models.

3.1 Uncertainty quantification

We illustrate uncertainty quantification with BART on the California Housing dataset, available in `sklearn`. Each observation is described by eight explanatory variables (such as median income, housing age, and population density), and the target variable is the median house value.

Posterior inference is performed using the posterior mean computed from 1000 samples after 1000 burn-in iterations, on 200 trees. Uncertainty intervals are then derived from the posterior distribution of the predictions. To study how uncertainty depends on the amount of data, we train BART on subsamples of 100, 1000, and 10000 observations. Results are available in appendix (Figure 3).

First, the training and test losses are relatively close, which illustrates the regularizing effect of the prior.

We quantify model uncertainty using 95% credible intervals. Note that these intervals quantify uncertainty on the regression function f rather than predictive uncertainty on y. The width of these intervals measures the epistemic uncertainty of the model. We compare this width to the scale of the observation noise. Since the noise is modeled as $\epsilon \sim \mathcal{N}(0, \sigma^2)$, a 95% interval for the noise

has approximate width 4σ. We therefore compare the width of the credible intervals for $f(x)$ to 4σ, where σ is estimated from the posterior.

We observe that the credible intervals for $f(x)$ are narrower than, but of the same order as, this noise scale. This indicates that the uncertainty on the estimated function is smaller than the intrinsic variability of the data, suggesting that the model captures the underlying signal reasonably well.

Moreover, this ratio remains relatively stable when increasing the size of the training dataset. We may partly explain this behavior by the regularizing effect of the prior, which constrains the complexity of the learned function and stabilizes the epistemic uncertainty.

Finally, in all three experiments, the trace of σ remains within a relatively narrow band around its posterior mean, indicating good convergence of the MCMC sampler after the burn-in phase.

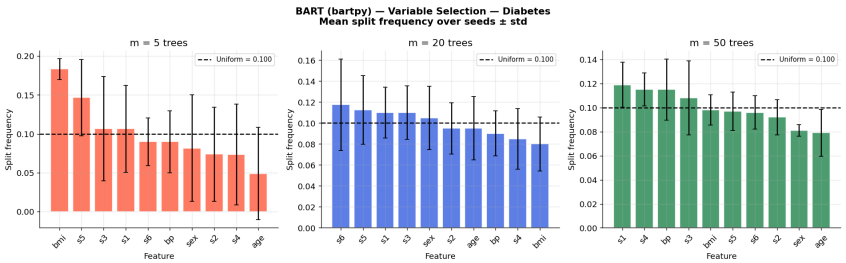
3.2 Variable selection

We study the ability of BART to perform variable selection using two indicators: *feature importance* and *partial dependence*. Feature importance measures how frequently a variable is used in tree splits, while partial dependence shows the marginal effect of a variable on the model prediction. Results are available on Figure 1 and Figure 4.

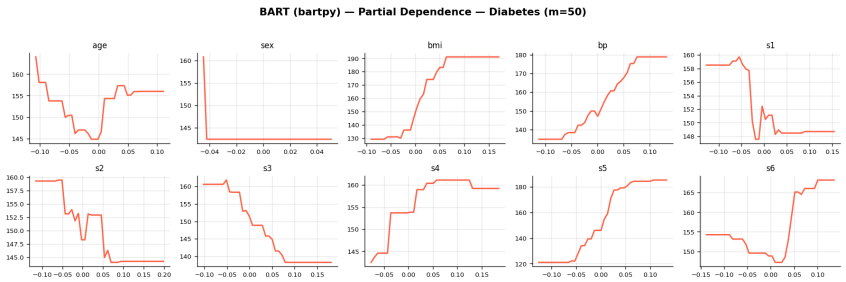
We evaluate this on two datasets. First, the diabetes dataset available in `sklearn`, a real-world regression dataset with ten clinical variables. Second, a simulated dataset generated with `make_friedman1`, which produces a nonlinear regression problem where only a subset of variables are relevant.

In both experiments, when the number of trees is relatively large (≥ 50), feature importances appear fairly uniform across variables. This behavior comes from the BART splitting mechanism: unlike Random Forests or Gradient Boosting, the splitting variable is proposed uniformly at random. As a result, many variables appear in splits even if they are not strongly informative. Only when the number of trees is very small (e.g., 5 trees on the diabetes data) do feature importances become informative.

In contrast, partial dependence plots clearly reveal the relevant variables. For the Friedman dataset, relevant variables exhibit a clear structure, while irrelevant variables produce curves resembling noise. A similar pattern is observed on the diabetes dataset.



(a) Feature importance



(b) Partial dependence

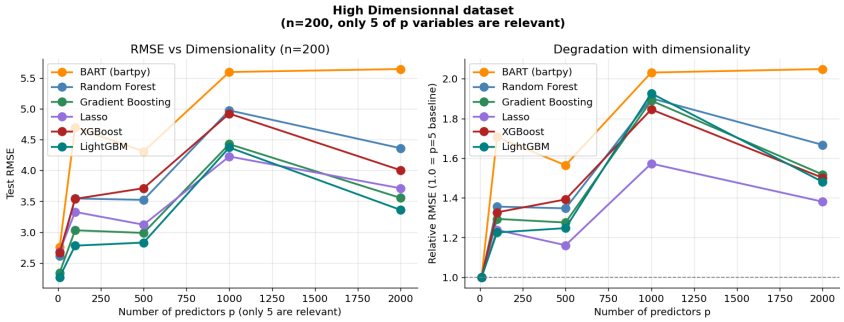
Figure 1: Diabetes dataset

3.3 High Dimensional Data

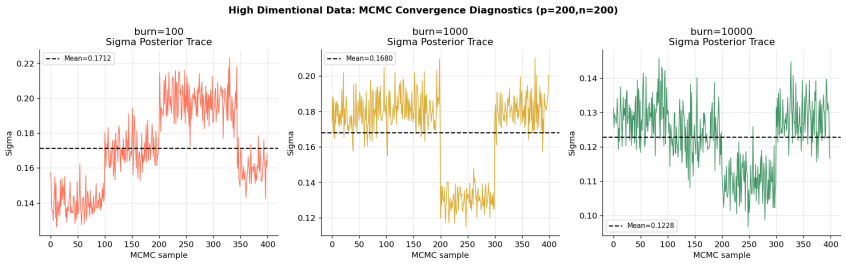
We use the Friedman1 dataset to study the behavior of BART in high-dimensional settings (see results in Figure 2). We observe that the performance of BART deteriorates as the number of uninformative variables increases, and becomes slightly worse than all baselines, although the gap remains moderate.

However, the posterior suggests slow mixing. In particular, the estimated noise standard deviation exhibits strong oscillations even with a relatively large burn-in period (about ten times larger than the one used in the previous section).

This slow mixing is likely due to the large space of trees that must be explored.. Indeed, the algorithm proposes splits by selecting variables uniformly at random, unlike standard boosting or bagging methods which choose splits based on an optimization criterion. As a result, in high-dimensional settings, many proposed splits are uninformative. This suggests that adapting the prior on split variables could improve mixing and scalability.



(a) Performance



(b) MCMC convergence

Figure 2: Friedman1 benchmark

4 Conclusion

BART is a Bayesian nonparametric supervised learning algorithm. It can be viewed as the Bayesian analogue of gradient boosting, where the prior on weak learners provides a regularizing effect. Unlike boosting or bagging methods, BART also allows one to quantify the epistemic uncertainty of the model.

However, the tree construction relies on uniformly proposed splits, which means that individual trees cannot be interpreted in the same way as those of Random Forests or Gradient Boosting. In particular, feature selection based on feature importance is limited, although we showed that partial dependence plots provide a more informative indicator.

Finally, while the method requires relatively little tuning of prior hyperparameters, it does require choosing the number of MCMC iterations. In high-dimensional settings, convergence can be slow, since feature exploration remains largely unguided compared to standard bagging or boosting approaches.

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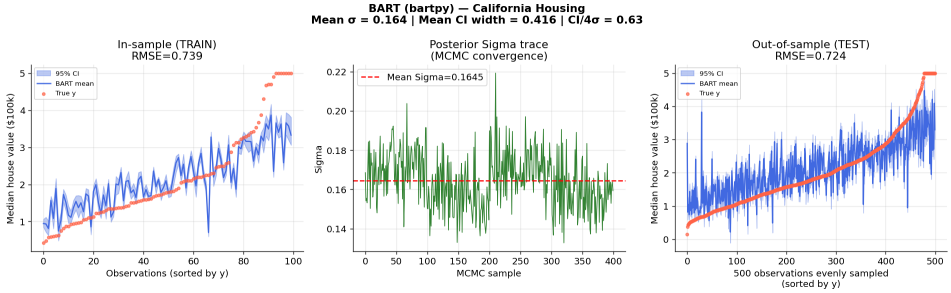
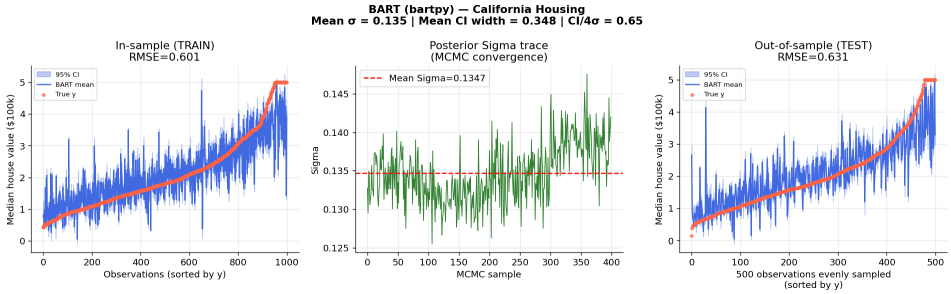
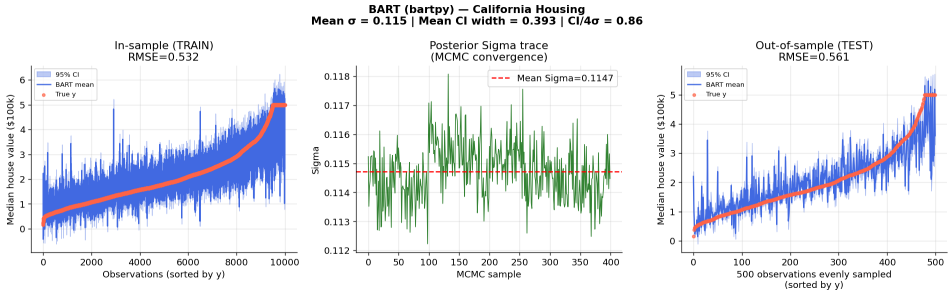
A California Housing experiment



(a) Training size = 100



(b) Training size = 1000



(c) Training size = 10000

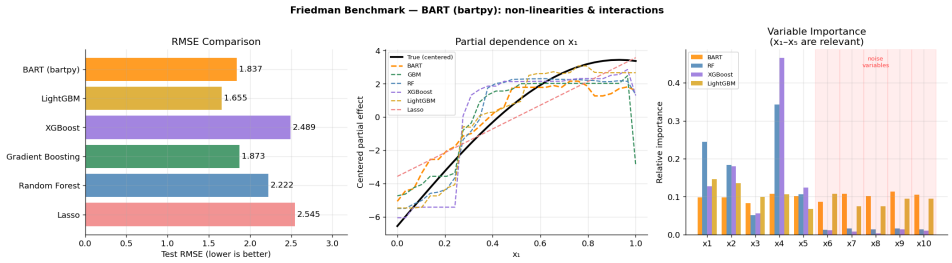
Figure 3: BART uncertainty analysis on the California Housing dataset for different training sizes.

B Friedman1 Benchmark experiment

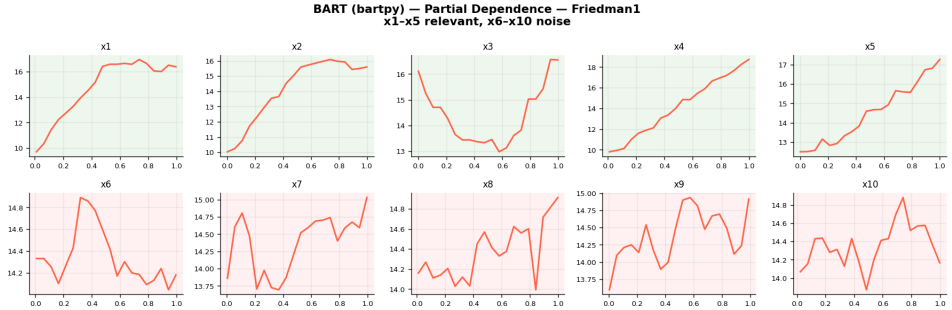
The Friedman1 benchmark is a synthetic nonlinear regression problem commonly used to evaluate feature selection methods. The target variable is generated as

$$y = 10\sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 + \epsilon,$$

where $x_i \sim \mathcal{U}(0,1)$ and $\epsilon \sim \mathcal{N}(0,1)$. Only the first five variables are relevant, while the remaining features are pure noise.



(a) Feature importance



(b) Partial dependence

Figure 4: Friedman1 benchmark